



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 6E: Shift and Rotate Operators

Operator Precedence

The following is the order of operator precedence

Class						
Highest	1. Miscellaneous op.	**	abs	not		
	2. Multiplying op.	*	/	mod	rem	
	3. Unary operators	+	-			
	4. Addition operators	+	-	&		
	5. Shift operators	sll	srl	rol	ror	
Lowest	6. Relational ops	=	/=	<	<=	> >=
	7. Logical operators	and	or	nand	nor	xor xnor

VHDL'93 Built-in Shift Functions 1

(numeric_std)

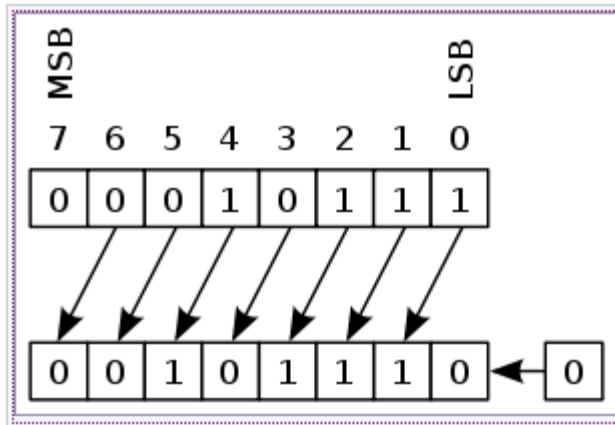
```
result <= operand1 operator operand2;  
           (left)           (right)
```



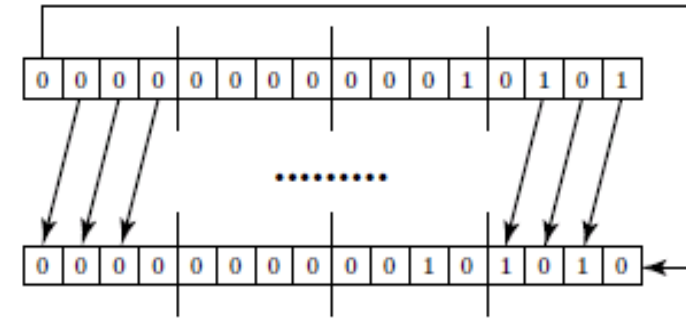
Operator	Description	Operand Type	Result Type
Shift sll	Shift left logical (fill right vacated bits with the 0)	Left: signed or unsigned Right: integer	Same as left type
Shift srl	Shift right logical (fill left vacated bits with 0)	same as above	Same as left type
Rotate rol	Rotate left (circular)	same as above	Same as left type
Rotate ror	Rotate right (circular)	same as above	Same as left type

VHDL'93 Built-in Shift Functions 2

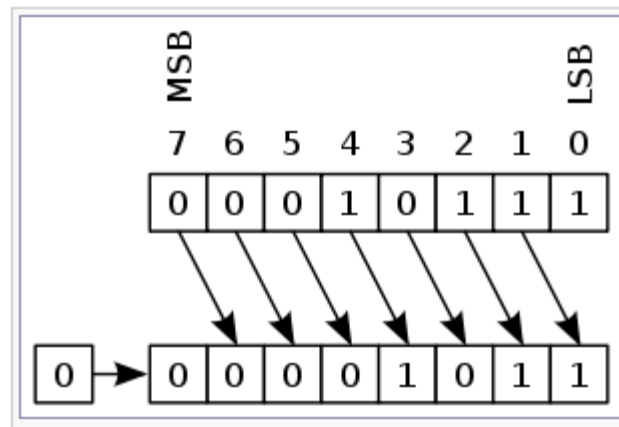
sll



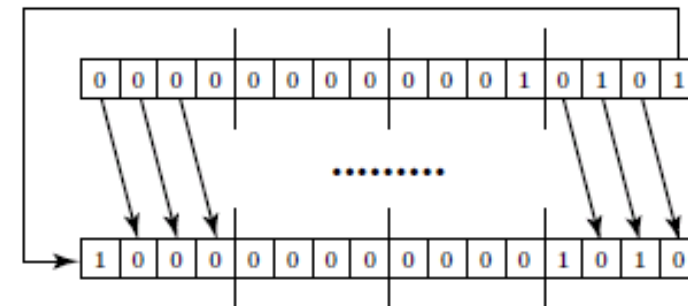
rol



srl



ror



VHDL'93 Built-in Shift Functions 3

- Example Shift operators:
- Let `A <= "10010101";` -- unsigned(7 downto 0)
 - shift left logical, filled with '0'
 - `A sll 2` results in `"01010100"`
 - shift right logical, filled with '0'
 - `A srl 3` results in `"00010010"`
 - rotate left by 3
 - `A rol 3` results in `"10101100"`
 - rotate right by -5
 - `A ror (-5)` results in `"10110010"`
- What if A was signed(7 downto 0)?